UNIVERSITÀ POLITECNICA DELLE MARCHE

2025 Projects

Luca Spalazzi
Dipartimento di Ingegneria dell'Informazione

CRISPY
CENTRO DI RICERCA E SERVIZIO PER LA PRIVACY E LA CYBERSECURITY

cini Cybersecurity National Lab

GRUPPO DI INGEGNERIA INFORMATICA - ING-INF/05

GII

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Tema 2025

Zero Trust Architecture

- Gruppi da 4-5 persone
- Usare Splunk, IpTables, Squid, Snort
- Usare un DBMS a scelta

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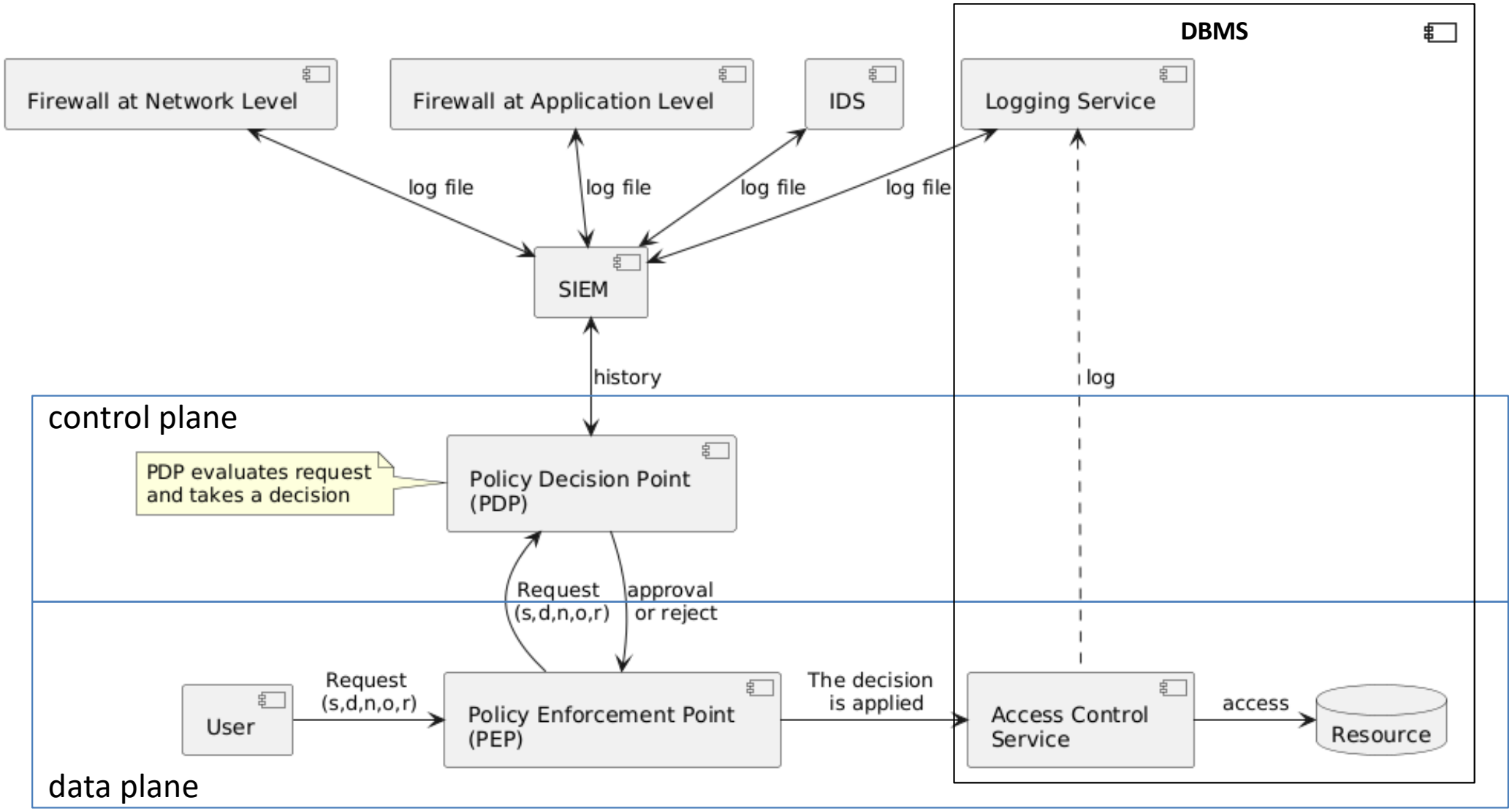
Zero Trust Architecture



Zero Trust Architecture

The next two slides report the sequence diagrams of the following two use cases:

1. resource access request (request that must be accepted by the PEP)
2. any other type of request (it is used to collect logs that integrate the history that must be analyzed by the PDP)

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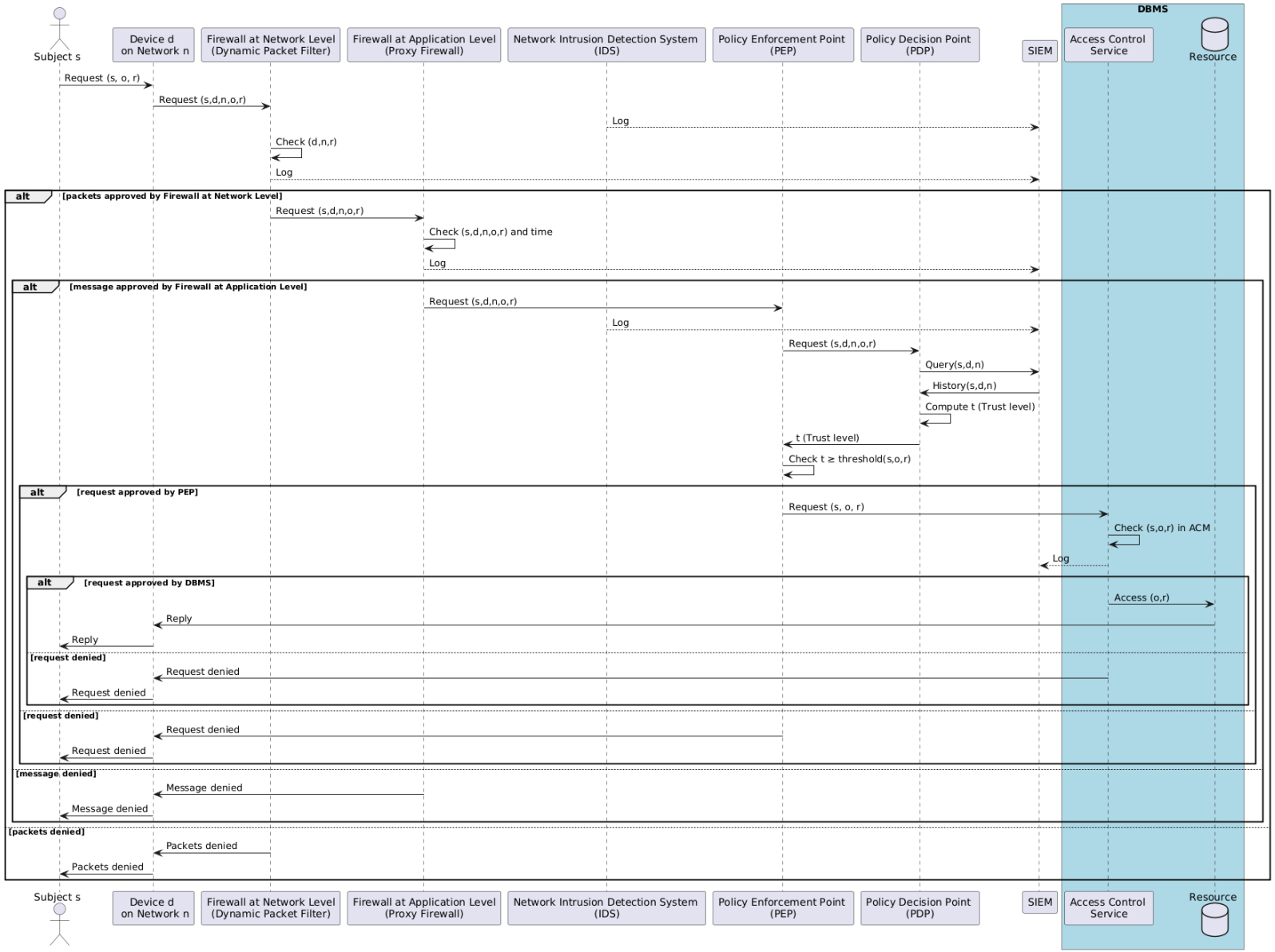
Zero Trust Architecture

Request access to the resource



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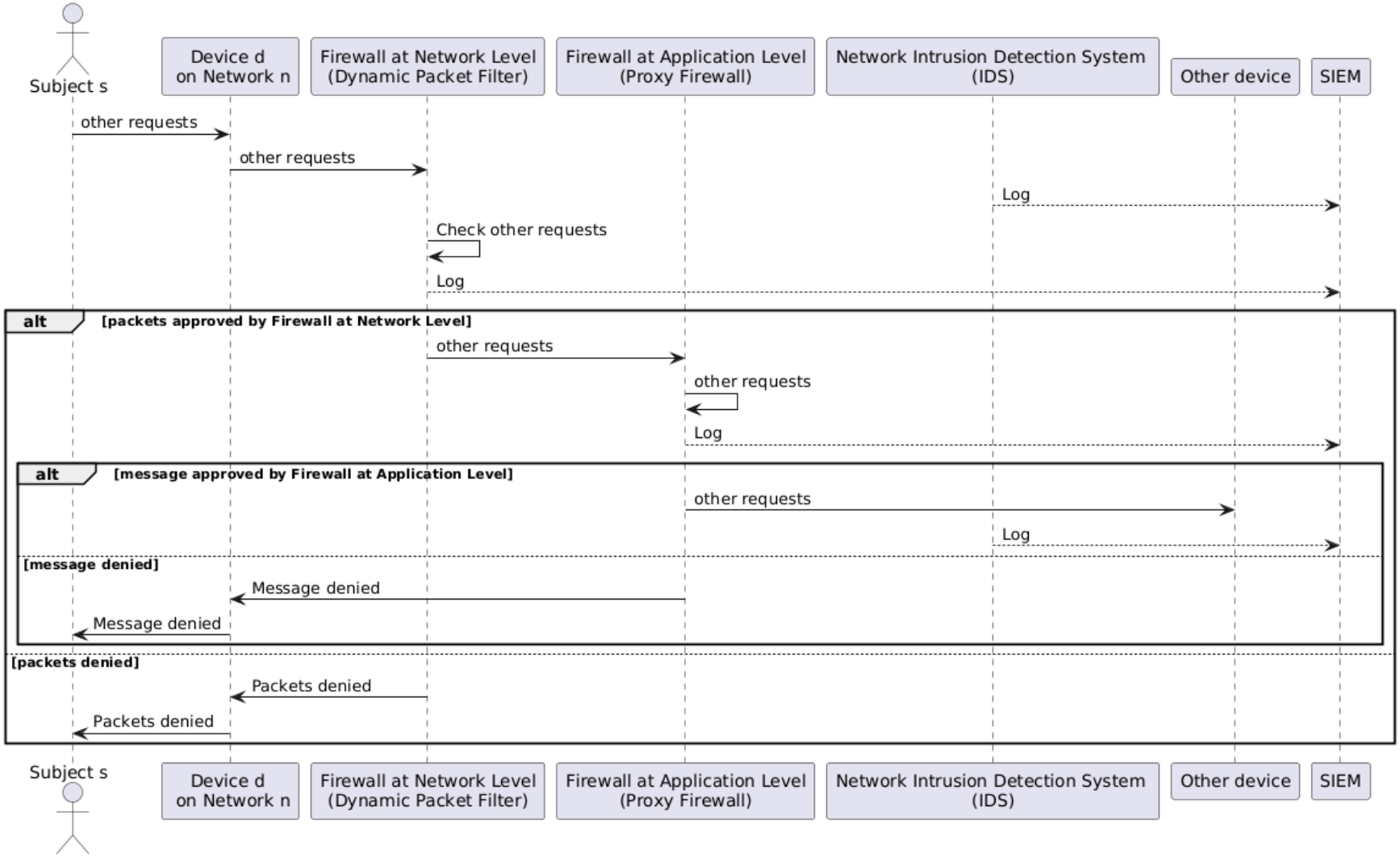
Zero Trust Architecture

Request access to something else



Tips and Tricks

- Packet Filter and Application-level firewall can identify the device ***d*** based on IP address and (limited to the same subnet) on MAC address
  - Notice that when traffic is routed between different networks, MAC addresses are changed
- Application-level firewall can identify the device ***d*** based on hostname.
- Packet Filter and Application-level firewall can identify the network ***n*** based on network ranges
- Application-level firewall can identify the network ***n*** based on top/second-level domains and subdomains
- PEP is a DBMS Client, i.e. PEP uses DBMS API
- PDP is a SIEM Client, i.e. PDP uses SIEM API

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Draft plan

1. Define policies
2. Install tools
3. Develop PDP and PEP
4. Test

QUESTIONS?